

3-Mark Questions & Answers

Q1. Differentiate between Data, Database, and DBMS.

Feature	Data	Database	DBMS
Meaning	Raw, unprocessed facts	Organized collection of related data	Software to manage the database
Example	`101`, `Alice`, `90%`	Student records table	MySQL, Oracle
Structure	No structure	Structured (tables, files)	System that imposes structure
Usefulness	Meaningless alone	Meaningful when queried	Makes database useful and accessible

Q2. List and briefly explain any three characteristics of a database.

1. **Data Integrity:** The DBMS enforces rules (constraints) to ensure that all data stored is accurate, valid, and consistent. For example, a student's age cannot be stored as a negative number.
2. **Data Security:** The DBMS controls who can access which data using authentication (login) and authorization (permissions/privileges). This prevents unauthorized access or modification.
3. **Reduced Data Redundancy:** In traditional file systems, the same data (e.g., a student's address) might be stored in multiple files, leading to inconsistency. A DBMS minimizes this by centralizing data and using normalization.

Q3. Explain the three levels of Data Abstraction.

The DBMS provides three levels of abstraction to hide complexity from users:

1. **Physical Level (Internal):** The lowest level. Describes **how** data is physically stored on the disk -- file formats, indexing, data blocks, and storage structures. Hidden from all users.
2. **Logical Level (Conceptual):** The middle level. Describes **what** data is stored and the relationships between them -- i.e., the table schemas and constraints. Used by DBAs and programmers.
3. **View Level (External):** The highest level. Describes only a **part** of the database as seen by a specific user or application. Different users may see different views. Provides simplicity and security.

View Level (What user sees -- partial view)
 Logical Level (What data exists -- full schema)
 Physical Level (How data is stored -- on disk)

Q4. Differentiate between Logical and Physical Data Independence.

Feature	Logical Data Independence	Physical Data Independence
Definition	Ability to change the logical schema w	Ability to change the physical schema
Level of Change	Conceptual/Logical schema changes	Physical/Internal schema changes
Example	Adding a new column to a table without	Changing from sequential file storage
Difficulty	Harder to achieve	Easier to achieve

Q5. Explain 1-tier, 2-tier, and 3-tier architecture in brief.

1. **1-Tier Architecture:** The client, application logic, and database all reside on one machine. Used for personal/local applications. Simple but not scalable. *Exam
2. **2-Tier Architecture:** The client (with application logic) communicates directly with a database server over a network. Better than 1-tier for small teams. *Exam
3. **3-Tier Architecture:** Three distinct layers -- (a) Presentation Tier (client UI), (b) Application/Logic Tier (business rules, middle-ware), and (c) Database Tier (data storage). Most scalable and secure; used in modern web applications.

Q6. What are the tasks/functions offered by a DBMS

1. **Data Definition:** The DBMS allows users to define the structure (schema) of the database -- creating tables, setting data types, and defining constraints using DDL (Data Definition Language).
2. **Data Updation:** The DBMS provides DML (Data Manipulation Language) commands like INSERT, UPDATE, and DELETE to add, modify, or remove data in the database.
3. **Data Retrieval:** Users can query the database to fetch specific data using the SELECT statement. The Query Processor optimizes and executes these queries.
4. **User Administration:** The DBA can create user accounts, grant or revoke privileges, and control access to specific tables or operations -- ensuring data security.

Q7. Explain any three components of the Storage Manager.

1. **Authorization and Integrity Manager:** Checks that a user has the required permissions before executing a data access operation. Also enforces integrity constraints (e.g., primary key, NOT NULL).
2. **Transaction Manager:** Ensures that database transactions execute reliably. It manages the ACID (Atomicity, Consistency, Isolation, Durability) properties, handles failure recovery, and maintains the database in a consistent state.
3. **Buffer Manager:** Manages the transfer of data between disk storage and main memory (RAM). It fetches data pages into a buffer pool in RAM and decides which pages to write back to disk, minimizing costly disk I/O operations.

Q8. Explain DDL interpreter, DML compiler, and Query evaluation engine in brief.

1. **DDL Interpreter:** Interprets Data Definition Language (DDL) statements like CREATE TABLE and DROP TABLE. It records the definitions (metadata) in the data dictionary.
2. **DML Compiler:** Translates DML queries (like SELECT, UPDATE) written in high-level SQL into low-level query plans (procedural instructions) that the query evaluation engine can execute.
3. **Query Evaluation Engine:** Executes the low-level query plan generated by the DML compiler. It coordinates with the Storage Manager to physically access the data on disk and return the result to the user.

Q9. List any three responsibilities of a DBA.

1. **Schema Definition:** The DBA writes DDL statements to create the original database schema (tables, views, indexes, constraints) and loads the initial data.
2. **Granting User Access:** The DBA creates user accounts and uses GRANT and REVOKE commands to control which users have access to which parts of the database, ensuring security.
3. **Routine Maintenance & Monitoring:** The DBA monitors system performance, performs periodic backups to prevent data loss, plans recovery procedures for failures, and tunes the database configuration for optimal performance.

Q10. Differentiate between Application Programmers and Sophisticated Users.

Feature	Application Programmers	Sophisticated Users
**Who are they **	IT professionals who develop database	Engineers, scientists, analysts
How they interact	Write programs using DML calls embedde	Write SQL queries directly or use tool
Output	Pre-built applications used by naive u	Ad-hoc queries and data analysis repor
Example	A developer who builds a library manag	A business analyst writing complex JOI

Q11. Explain the need for a database (any three points).

1. **Eliminating Data Redundancy and Inconsistency:** In file systems, the same data (e.g., a customer's phone number) is stored in multiple files, leading to inconsistency when one is updated but others are not. A DBMS stores data centrally, eliminating redundancy.
2. **Difficulty in Accessing Data:** Retrieving data from flat files requires writing custom programs for every query. A DBMS provides a powerful query language (SQL) that allows easy, ad-hoc retrieval of data without programming.

3. Data Security and Integrity: File systems have limited access control. A DBMS provides fine-grained security using user accounts, privileges, and integrity constraints (like NOT NULL, FOREIGN KEY), ensuring only valid and authorized data operations occur.

Q12. Write short notes on Specialized Users.

Specialized Users are sophisticated users who write specialized database applications that don't fit the traditional data processing framework. They work with advanced database features and technologies such as:

* Kn

Q13. Differentiate File System and Database approach (any 3 points).

Feature	File System	Database System
Data Redundancy	High -- same data stored in multiple f	Low -- data stored centrally, redundan
Data Access	Requires custom programs to access/que	Standard SQL provides easy, powerful,
Data Security	Limited security; access is mainly at	Fine-grained security with user accoun
Data Integrity	No built-in constraints; integrity is	DBMS enforces constraints (PK, FK, NOT
Concurrency	Very limited support; simultaneous fil	Built-in concurrency control ensures s

Q14. Explain Entity, Entity Set, and Attributes with an example.

* Entity: A specific, distinguishable object or concept in the real world. Example: Student "Suraj" with ID 101.

* En

Q15. Explain the components of a Relationship Set with an example.

A Relationship Set is a mathematical relation among ≥ 2 entity sets.

* Pa

Q16. Differentiate between Simple and Composite attributes with examples.

Feature	Simple Attribute	Composite Attribute
Definition	Atomic; cannot be divided into smaller	Can be divided into smaller sub-compon
Structure	Single component	Multi-component hierarchy
Example	Age, Salary, Roll_No	Name (First_Name, Last_Name), Address
Query Use	Queried directly	Sub-parts can be queried individually

Q17. Differentiate between Single-valued and Multivalued attributes.

Feature	Single-Valued Attribute	Multivalued Attribute
Values per Entity	Holds exactly one value per entity ins	Can hold zero, one, or multiple values
ER Symbol	Single Oval / Ellipse	Double Oval / Ellipse
Example	Date_of_Birth, SSN	Phone_Number (work, home, mobile), Deg
Relational Mapping	Map directly as a column in the main e	Requires a separate table with primary

Q18. Explain Derived Attributes with an example.

A Derived Attribute is an attribute whose value is not physically stored in the database but is computed dynamically on demand from other stored attributes or system functions.

* ER

Q19. Explain the ER diagram notation for entity sets, attributes, and relationships (symbols used).

Rectan

Double Ellipse Dashed Ellipse Diamond

/=====

Q20. Explain the four types of mapping cardinality with examples.

1. One-to-One (1:1): An entity in A is associated with at most one entity in B, and vice versa. Example: EMPLOYEE *manages* DEPARTMENT (each employee manages at most 1 department; each department is managed by 1 employee).

2. On

A. Exa

Q21. Differentiate between Total and Partial Participation with examples.

Feature	Total Participation	Partial Participation
Constraint	Every entity in the set must parti	Some entities may not participate
Dependency	Existence dependency (mandatory)	Optional participation
ER Line Symbol	Double Line	Single Line
Example	Every LOAN must belong to a CUSTOMER	Not every EMPLOYEE manages a DEPARTM

Q22. Explain weak entity set and strong entity set with an example.

* Strong Entity Set: Has a primary key that uniquely identifies every entity. Does not depend on any other entity set. Example: EMPLOYEE(Emp_ID, Name, Salary).

* W

(partial

Q23. Explain how a primary key is derived for a relationship set (1:1, 1:N, M:N).

The primary key of a relationship set depends on its mapping cardinality:

* Ma

Q24. Explain Role in a relationship with an example (e.g., prerequisite).

A Role is the explicit label given to an entity set's participation in a relationship set when the same entity set participates more than once (recursive relationship).

* Ex

Q25. Explain Overlapping and Disjoint Specialization with examples.

* Disjoint Specialization: An entity can belong to at most one lower-level subclass. Represented by 'd' in ER circle. Example: EMPLOYEE is either HOURLY_EMPLOYEE OR SALARIED_EMPLOYEE (cannot be both).

* C
Examp

Q26. Explain the Completeness Constraint (Total/Partial generalization).

The Completeness Constraint specifies whether every entity in the superclass MUST belong to at least one subclass:

* To
circle).

Q27. Explain Attribute Inheritance in generalization/specialization.

Attribute Inheritance is a fundamental property of ISA hierarchies.

* Ev

Q28. Explain Condition-defined vs User-defined lower-level entity sets.

* Condition-Defined (Predicate-Defined): Membership in a subclass is automatically determined by an explicit condition/predicate on an attribute of the superclass. Example: If Job_Type = 'Secretary', entity automatically belongs to SECRETARY subclass.

* U
Examp

Q29. What is a redundant attribute Explain with the Student-Department example.

A Redundant Attribute is an attribute that introduces duplicate information that can already be derived from existing relationships or other tables.

* E
Dept_I

Q30. Explain Key Constraint on Entity Set and Relationship Set.

* Key Constraint on Entity Set: Ensures every entity has a unique identifier (Primary Key / Candidate Key) so no two entities share identical key values.

* K
between

Q31. Identify the entity sets and relationships for the Hospital Management System, and classify each relationship as 1:1, 1:N, or M:N.

* Entity Sets: PATIENT, DOCTOR, ROOM, MEDICINE.

* Re

Q32. Explain the Foreign Key approach for mapping a binary 1:1 relationship, using EMPLOYEE-DEPARTMENT (Manages).

To map a binary 1:1 relationship MANAGES between EMPLOYEE and DEPARTMENT:

1. Cr

Q33. For the Online Shopping System, describe the mapping of "Customer places Order" (which side gets the foreign key, and why).

- * Relationship: CUSTOMER *places* ORDER is a 1:N relationship (1 Customer places N Orders; 1 Order belongs to 1 Customer).

* Ma

Q34. Explain how multivalued attributes are mapped to a relational schema, with an example (e.g., DEPT_LOCATIONS).

- * Mapping Rule: A multivalued attribute cannot be stored directly in a relational table (violates 1NF). Create a new separate relation containing:

1. Th

Q35. Differentiate "Multiple relations - Superclass and subclasses" vs "Single relation with one type attribute" for specialization mapping.

Feature	Option 1: Multiple Relations (Super +	Option 3: Single Relation (One Type At
Tables Created	1 table for superclass + 1 table per s	Exactly 1 single unified table
Handling Subclass Attributes	Subclass tables store only specific at	Single table holds ALL attributes for
NULL Values	Zero NULL values	Many NULL values for non-applicable su
Best Used When	Subclasses have many specific attribut	Specialization is disjoint with few sp

Q36. For the Movie Ticket Booking System, identify entity sets and the cardinality of each relationship.

- * Entity Sets: USER, THEATRE, SHOW, TICKET.

* Re

Q37. Explain why an M:N relationship set requires a new relation, unlike 1:N relationships.

- * In 1:N relationships: The "N" side entity participates in at most 1 relationship instance. Hence, we can cleanly place the "1" side's primary key as a Foreign Key directly inside the "N" side table without violating 1NF.

* In

Q38. What are the three approaches for mapping a binary 1:1 relationship When is each appropriate

1. Foreign Key Approach: Add Foreign Key of one relation into the other. *Best when one entity set has total participation (place FK in total side).*

2. M

